

Experiment 5: K-Nearest Neighbours (K-NN)

Aim

To implement the **K-Nearest Neighbours (K-NN)** algorithm in Python for classification.

Algorithm

Step 1: Import the required libraries.

Step 2: Load the dataset.

Step 3: Separate the features and target variable.

Step 4: Split the dataset into training and testing data.

Step 5: Create a K-NN classifier with a suitable value of K.

Step 6: Train the classifier using the training data.

Step 7: Predict the classes for the test data.

Step 8: Calculate and display the accuracy.

Program

```
# Experiment 5 - K-Nearest Neighbours

import pandas as pd
from sklearn.model_selection import train_test_split
from sklearn.neighbors import KNeighborsClassifier
from sklearn.metrics import accuracy_score

data = pd.read_csv("Iris.csv")

X = data.iloc[:, :-1]
y = data.iloc[:, -1]

X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.2, random_state=42
)

model = KNeighborsClassifier(n_neighbors=3)
model.fit(X_train, y_train)

y_pred = model.predict(X_test)

print("Predictions:", y_pred)
print("Accuracy:", accuracy_score(y_test, y_pred))
```

Output

```
*** Predictions: ['Iris-versicolor' 'Iris-setosa' 'Iris-virginica' 'Iris-versicolor'
'Iris-versicolor' 'Iris-setosa' 'Iris-versicolor' 'Iris-virginica'
'Iris-versicolor' 'Iris-versicolor' 'Iris-virginica' 'Iris-setosa'
'Iris-setosa' 'Iris-setosa' 'Iris-setosa' 'Iris-versicolor'
'Iris-virginica' 'Iris-versicolor' 'Iris-versicolor' 'Iris-virginica'
'Iris-setosa' 'Iris-virginica' 'Iris-setosa' 'Iris-virginica'
'Iris-virginica' 'Iris-virginica' 'Iris-virginica' 'Iris-virginica'
'Iris-setosa' 'Iris-setosa']
Accuracy: 1.0
```

Result

Thus, the K-Nearest Neighbours algorithm was successfully implemented in Python and the test data was classified successfully.